

Computational Methods in Econometrics

Investigating Temperature Trends: From Linear Trends to Structural Change

Assignment

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Presentation deadline: **06.10.2026 18:00**

Group size: **4 students**

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Overview

Instrumental temperature records are an important source of evidence on long-term climate change. Estimating and interpreting temperature trends, however, requires more than fitting a regression line. Residual autocorrelation, heteroskedasticity, and structural changes can affect statistical inference and the conclusions drawn from an empirical analysis.

In this assignment, each group analyzes annual temperature observations from one weather station. You will first estimate a linear trend and evaluate its residual properties. You will then investigate whether the trend changes over time, using bootstrap methods for inference. Finally, you will conduct a simulation study to compare the finite-sample performance of alternative bootstrap procedures.

Throughout this assignment, keep in mind the central research question: What does the temperature record at your assigned station reveal about long-term warming, model adequacy, and possible structural change?

Data

Each group will receive a separate annual temperature series from the same underlying station data source, NOAA National Centers for Environmental Information. The supplied data file will contain at least the calendar year, annual mean temperature, station identification, and, whenever available, basic station metadata.

Use the complete sample supplied for your station and clearly document any transformations you make to your data.

Assignment structure

The assignment is designed as one coherent empirical research project. Your report should therefore explain and justify the choices made throughout the analysis, rather than merely presenting a sequence of numerical answers.

The econometric and statistical content consists of three parts. The first part corresponds to the first topic of the course, Monte Carlo methods. The second part evolves around bootstrapping, while the third part combines both. At the end of the course, the main results need to be summarized in a presentation which will be followed by a Q&A.

1. Linear trend analysis and model diagnostics

Assess whether a simple linear trend provides an adequate initial description of the annual temperature series. At this stage, focus on estimation, interpretation, and diagnostics. Do not yet draw conclusions about the statistical significance of the estimated trend.

Consider the model

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t, \quad t = 1, \dots, n, \quad (1)$$

where y_t denotes annual mean temperature and t is a deterministic time index. You may centre or rescale the time index if this improves interpretation; if you do so, explain the transformation.

1. **Explore the data.** Produce an informative plot of the annual temperature series and describe its main empirical features. Comment on the apparent direction of the trend, interannual variability, unusual observations, and any visible changes in the behaviour of the series.
2. **Estimate the linear trend.** Estimate model (1) by OLS. Report and interpret the estimated intercept and slope. Express the trend in a scientifically meaningful unit, such as degrees Celsius per decade.

3. **Inspect the residuals.** Use appropriate residual plots and graphical diagnostics. For example, you can include:

- residuals plotted over time,
- a residual autocorrelation function,
- a plot suitable for assessing changes in residual variance,
- a normal QQ plot.

Explain what each diagnostic suggests.

4. **Test for first-order residual autocorrelation.** Apply the Durbin–Watson test to the residuals from model (1). Use the statistic

$$\hat{d} = \frac{\sum_{t=2}^n (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^n \hat{\varepsilon}_t^2}. \quad (2)$$

(a) Construct a two-sided Monte Carlo test with $B = 9,999$ simulations and significance level $\alpha = 0.05$. Because the finite-sample null distribution need not be symmetric, obtain separate lower and upper critical values. What does the rejection region look like?

(b) Report $\hat{d}$, the Monte Carlo critical values, the Monte Carlo p -value, and your conclusion.

5. **Test for heteroskedasticity.** Consider the auxiliary regression

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 t + v_t. \quad (3)$$

If you centred or rescaled the time index in model (1), use the corresponding transformed trend regressor in the auxiliary regression. Test

$$H_0 : \gamma_1 = 0 \quad \text{vs.} \quad H_1 : \gamma_1 \neq 0.$$

Use the Breusch–Pagan statistic

$$BP = nR^2, \quad (4)$$

where R^2 is obtained from the auxiliary regression (3). Under H_0 , BP has an asymptotic χ_1^2 distribution. Apply the Breusch–Pagan test to the residuals from (1) and report the value of BP , the corresponding p -value, and your empirical conclusion.

6. **Compare the inferential principles.** Explain why it is appropriate to construct Monte Carlo critical values for the Durbin–Watson statistic using the procedure above, while the same principle cannot simply be transferred to the finite-sample Breusch–Pagan statistic.

7. **Assess the initial model.** Bring the graphical and formal diagnostics together.

2. Structural change and bootstrap inference

In this part, you will investigate whether the temperature trend changes at an unknown date. Implement a test for a break in the trend, estimate the break date, and use bootstrap methods for both hypothesis testing and confidence intervals. Your choice of bootstrap procedure should be in line with the residual diagnostics obtained in Part I.

The broken-trend model

For this part, consider the continuous broken-linear-trend model

$$y_t = \alpha + \beta t + \delta D_{t,T_1} + \varepsilon_t, \quad t = 1, \dots, T, \quad (5)$$

where

$$D_{t,T_1} = \begin{cases} 0, & \text{if } t \leq T_1, \\ t - T_1, & \text{if } t > T_1, \end{cases} \quad (6)$$

and T_1 denotes the unknown break date. The intercept and pre-break slope are α and β , respectively. After the break, the slope equals $\beta + \delta$. The specification changes the slope while enforcing continuity at the break date.

Let Λ denote the set of admissible candidate break dates. For a trimming fraction $0 < \lambda < 1/2$, define

$$\Lambda = [\lambda T, (1 - \lambda)T], \quad (7)$$

interpreted as the corresponding set of integer-valued time indices. Unless stated otherwise, use $\lambda = 0.15$.

1. **Understand and estimate the model.** Explain how model (5) differs from the simple linear trend model. For a fixed candidate break date $T_c \in \Lambda$, explain how the parameters can be estimated by OLS.
2. **Estimate the break location and fit the broken-trend model.** Estimate model (5) for every candidate break date $T_c \in \Lambda$ and record the corresponding residual sum of squares,

$$RSS(T_c) = \min_{\alpha, \beta, \delta} \sum_{t=1}^T (y_t - \alpha - \beta t - \delta D_{t,T_c})^2.$$

Estimate the break date by

$$\hat{T}_1 = \arg \min_{T_c \in \Lambda} RSS(T_c). \quad (8)$$

Report the estimated calendar year. Include a plot of $RSS(T_c)$, or equivalently the reduction in residual sum of squares relative to the no-break model, against the candidate break year. Estimate model (5) at $\hat{T}_1$ and plot the fitted broken-trend line together with the observed data. At this stage, treat $\hat{T}_1$ as the best-fitting break location; do not yet interpret it as evidence that a structural break is present.

3. **Construct the observed structural break statistic.** Test

$$H_0 : \delta = 0 \quad \text{against} \quad H_1 : \delta \neq 0 \text{ for some } T_c \in \Lambda.$$

Let RSS_0 denote the residual sum of squares from the no-break model. Define

$$F_T = RSS_0 - RSS(\hat{T}_1). \quad (9)$$

This is equivalent to

$$F_T = \min_{\alpha, \beta} \sum_{t=1}^T (y_t - \alpha - \beta t)^2 - \inf_{T_c \in \Lambda} \min_{\alpha, \beta, \delta} \sum_{t=1}^T (y_t - \alpha - \beta t - \delta D_{t,T_c})^2.$$

Large values of F_T indicate stronger evidence in favour of the broken-trend model. Implement the procedure yourself rather than using a function that directly returns the complete unknown-break test. Report the observed value of F_T .

4. **Choose bootstrap procedures.** Motivate the use of bootstrap inference. Based on the residual diagnostics in Part I, select one *primary* bootstrap procedure that you regard as appropriate for the station data and at least one additional procedure for sensitivity analysis.
5. **Write down the algorithms.** Before presenting results, state the complete bootstrap algorithm for every procedure used in this part. Your description must make clear:
 - which model is estimated under the null;
 - which residuals or innovations are resampled;
 - whether and how residuals are centred or rescaled;
 - how each bootstrap sample $y_1^*, \dots, y_T^*$ is generated;
 - how F_T^* , the bootstrap critical value, and the bootstrap p -value are calculated;
 - how dependence and/or heteroskedasticity is preserved.
6. **Bootstrap the break test.** Use at least $B = 999$ bootstrap replications for the final reported results. Compare the conclusions across procedures. Based on the bootstrap test, assess the evidence for a structural break. If the no-break null is not rejected, explain how this affects the interpretation of the estimated break date and the fitted broken-trend model.
7. **Estimate and interpret the broken trend.** Using model (5) estimated at $\hat{T}_1$, report and interpret:
 - the estimated pre-break trend $\hat{\beta}$;
 - the estimated change in trend $\hat{\delta}$;
 - the estimated post-break trend $\hat{\beta} + \hat{\delta}$.

Express slopes in degrees Celsius per decade. Interpret these quantities in light of the evidence from the bootstrap break test, keeping the best-fitting broken-trend specification distinct from statistical evidence that a break is present.

8. **Construct bootstrap confidence intervals.** For the pre-break and post-break slopes, construct at least two types of bootstrap confidence intervals, including an equal-tailed percentile interval and an equal-tailed percentile- t interval.
Use your primary bootstrap procedure and re-estimate the break date in each bootstrap replication. Explain any additional studentisation required for the percentile- t interval. State whether the intervals contain zero, while keeping the statistical and substantive interpretations distinct.
9. **Compare the models.** Compare the simple linear trend and broken-trend models using fitted plots, residual diagnostics, and an appropriate measure of fit. Discuss whether the broken-trend specification provides a substantively and statistically more convincing description of the station record.

3. Simulation study of bootstrap procedures

Evaluate the finite-sample performance of bootstrap methods in a trend or broken-trend setting. The simulation should help determine when the procedures used in Part II provide reliable inference and when they do not.

Base the simulation on

$$y_t = \alpha + \beta t + \delta D_{t,T_1} + \varepsilon_t, \quad (10)$$

$$\varepsilon_t = \rho \varepsilon_{t-1} + \sigma_t u_t, \quad u_t \stackrel{\text{iid}}{\sim} F_u, \quad (11)$$

where F_u has mean zero and variance one. The scale σ_t may be constant or time-varying. You must use this DGP family, but your group has discretion over a subset of design choices.

1. **Select meaningful scenarios.** Include at least one no-break scenario ($\delta = 0$) for size and at least one break scenario ($\delta \neq 0$) for power. In addition, choose at least two of:
 - weak versus strong change in trend;
 - independent versus autocorrelated errors;
 - homoskedastic versus heteroskedastic errors;
 - central versus early or late break;
 - shorter versus longer sample size.

Whenever possible and useful, calibrate the sample size and error magnitude to the empirical station data.

2. **Compare bootstrap methods.** Compare at least **three distinct bootstrap procedures**. Include at least one method designed for independent errors, one capable of preserving serial dependence, and, whenever a heteroskedastic DGP is included, one appropriate under heteroskedasticity. Depth is valued more than the number of algorithms.
3. **Implement the Monte Carlo experiment.** Use at least $M = 1,000$ Monte Carlo replications per configuration and at least $B = 499$ bootstrap replications within each Monte Carlo replication, unless computational constraints are documented. Set and report a random seed.

Remark. The simulation design is deliberately partly open. Credit is awarded for a coherent design, correct implementation, and insightful interpretation. Adding many scenarios without explaining what they teach will not improve the assessment.

Report requirements

Submit one report per group. It should read as a short empirical research paper rather than a list of isolated answers. A suitable structure is:

1. Introduction and research question
2. Data and exploratory analysis
3. Linear trend and model diagnostics
4. Structural change and bootstrap inference
5. Simulation study of bootstrap procedures
6. Conclusion.

The main text should not exceed 15 pages, excluding the title page, references, and appendix. Place essential results in the main text. Long algorithm listings, supplementary simulation tables, and robustness checks may be placed in an appendix. All figures must display meaningful axis labels, units, and captions.

Submit all code in a separate, fully reproducible code file. It should run from a clean session using relative file paths. Do not submit output that cannot be reproduced from the supplied code and data.

Final presentation and presentation Q&A

Each group will record a short presentation on its assigned station and upload it on Canvas.

- Presentation length: 10 minutes, every team member has to present a part.
- Focus on the scientific question, methodological choices, and main findings.
- Do not attempt to present every result.
- During the Q&A, you are asked to start with a self-reflection on your work (2 minutes), followed by approx. 10 minutes of discussion and questions.

Assessment criteria

The provisional marking scheme is:

Component	Weight
Part I	20%
Part II	25%
Part III	25%
Final presentation	15%
Presentation Q&A	15%
Total	100%

Academic integrity and use of software and AI

You may use standard functions for elementary estimation, plotting, matrix operations, and auxiliary regressions. Unless explicitly permitted, you may not use a package function that performs an entire required Monte Carlo test, unknown-break test, or bootstrap analysis as a single black-box command.

The use of AI tools and large language models (e.g., ChatGPT, Claude, Copilot) is permitted only for:

- debugging code that you have written yourself,
- improving the grammar, spelling, or clarity of your writing.

AI tools may not be used to generate econometric analyses, interpret results, write substantial parts of the report, or produce code or solutions for the assignment. All submitted work must reflect your own understanding and be your own work. Clearly acknowledge all external sources, packages, and whether and for what you have made use of AI and LLMs. Every group member is responsible for understanding all submitted code and analysis.

Good luck!